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Key Learnings & Challenges

Building this end-to-end classification pipeline highlighted the critical importance of preventing data leakage during preprocessing. Isolating feature scaling, imputation, and SMOTE exclusively to the training split ensured that our validation and test metrics reflected realistic performance.

Technical Takeaways

Feature Engineering Matters: Adding meteorological proxies (diurnal range, soil-air temp difference) directly boosted model F1-scores across tree models.

Ensembling Advantages: Combining prediction probabilities from XGBoost, Random Forest, and Support Vector Machines via Bayesian weighting systematically outperformed every individual base classifier.